Data Import Description

The data used for the Business Database is synthetically generated using an AI system. The exact generation prompt is documented in the following section

The generated dataset represents realistic bank card transaction data and contains a limited number of intentionally introduced data-quality issues, such as missing optional values, inconsistent capitalization and whitespace, different date and decimal formats, duplicate records and invalid values.

The source data is provided as T-SQL INSERT statements and initially loaded as raw data into staging tables within the stg schema using text-based fields. The data is then validated, cleaned and converted into the required data types using T-SQL transformations such as `TRIM`, `REPLACE` and `TRY_CONVERT` before being inserted into the normalized Business Database.

The dataset contains approximately **120 customers, 140 accounts, 180 cards, 40 merchants, 12 merchant categories, 155 ownership relationships and 2,000 transactions**.

The corresponding staging, INSERT and transformation scripts are provided separately with the project files.

Data Generation Prompt

Using GPT-5.6 Sol

> Create a synthetic source dataset for a training Data Engineering project called Bank Card Transaction Analysis.
>
> The target Business Database consists of the following relations:
>
> `Customer(CustomerID, FirstName, LastName, BirthDate, Email, Phone, Street, PostalCode, City, Region, Country)`
>
> `Account(AccountNumber, Type, Currency, Status, OpeningDate)`
>
> `Card(CardNumber, ExpiryDate, Type, Status, CustomerID, AccountNumber)`
>
> `Transaction(TransactionID, Date, Time, Status, Type, Currency, Amount, CardNumber, MerchantID)`
>
> `Merchant(MerchantID, Name, Street, PostalCode, City, Region, Country, MerchantCategoryID)`
>
> `MerchantCategory(MerchantCategoryID, CategoryName, Description)`
>
> `AccountOwnership(CustomerID, AccountNumber, Role)`
>
> Generate approximately:
>
> 120 customers, 140 accounts, 180 cards, 40 merchants, 12 merchant categories, 155 ownership relationships and 2,000 transactions.
>
> The generated data should cover approximately **2–3 years** so that monthly, quarterly, seasonal and yearly transaction analysis is meaningful.
>
> Preserve the logical business relationships. A card must belong to one customer and one account, and the cardholder must be an owner of that account. Some accounts should be joint accounts with two owners. Transactions must reference existing cards and merchants.
>
> The source data must **not be perfectly clean**. Introduce a realistic and limited amount of data-quality problems. Most records should remain valid. Include, in appropriate proportions:

> - missing values only where realistically possible, especially optional customer contact information;
> - leading/trailing whitespace;
> - inconsistent capitalization;
> - variations in country, region, category and status spelling;
> - different telephone-number formats;
> - different date formats;
> - decimal numbers using both `.` and `,`;
> - a small number of duplicate or near-duplicate source records;
> - a very small number of invalid values that should be rejected or corrected during cleansing.

Do **not** corrupt the majority of the dataset and do not introduce errors randomly into every table. The data-quality issues should resemble realistic operational source-data problems.

Use staging tables inside the SQL Server schema `stg` (for example `stg.Customer`, `stg.Transaction`). The generated INSERT statements must insert the raw, intentionally imperfect values into these staging tables, not directly into the final Business Database tables.

Output the data as **T-SQL INSERT statements for raw staging tables**, not as INSERT statements directly into the final typed Business Database tables. Assume staging attributes can initially be stored as `NVARCHAR` so that malformed source values can be imported without failing.

Split large transaction inserts into manageable batches.

Do not clean or normalize the erroneous source values in advance. The purpose is to import the raw source and subsequently demonstrate cleansing and transformation in SQL.

Also provide a short summary specifying which types of intentional data-quality issues were introduced and approximately how many records are affected by each issue.

Follow-up Correction Prompt

After reviewing the generated dataset, a few values were identified that did not fully match the defined project scope, particularly transaction types related to cash withdrawals and card verification. A follow-up prompt was therefore used to correct these scope inconsistencies and update the renamed `AccountOwnership` staging table without regenerating the dataset from scratch.

> Modify the existing generated dataset only; do not regenerate it from scratch.
>
> Replace all references to `[stg].[Owns]` with `[stg].[AccountOwnership]`.
>
> The project covers **bank card payment transactions only**. Transaction types must therefore represent events related to merchant card payments. Use the canonical transaction types `Purchase`, `Refund`, `Reversal`, and `Chargeback`. Intentional capitalization, whitespace, or minor spelling variations of these values may remain as data-quality issues.
>
> Remove/replace all semantically out-of-scope transaction types, especially `Cash Withdrawal`, `ATM Withdrawal`, and `Card Verification`. Do not introduce bank transfers, cash advances, deposits, or other non-payment banking operations.
>
> Use the following canonical transaction statuses: `Completed`, `Pending`, `Declined`, `Reversed`. Small intentional formatting/spelling variations may remain.
>
> The 12 canonical merchant categories are: `Groceries & Supermarkets`, `Restaurants & Cafes`, `Fuel & Automotive`, `Travel & Transport`, `Hotels & Lodging`, `Retail & Fashion`, `Electronics`, `Health & Pharmacy`, `Entertainment`, `Utilities & Telecom`, `Education`, and `Online Services`. Existing intentional capitalization, spelling, and whitespace problems may remain in the raw staging data.
>
> Preserve the existing record counts, IDs, dates, foreign-key relationships, transaction distribution, and intentional data-quality issues wherever possible. Only make the corrections described above.